

STOCHASTIC INTEGRALS

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I TO PROCESS INFORMATION

In Article No.31, we looked at Brownian motion as a mathematical model for continuous random movement. In this article we will advance and introduce stochastic integrals and Ito processes, and in particular, explore Ito's lemma, the stochastic analogue of the chain rule from ordinary calculus.

STOCHASTIC INTEGRALS

The stochastic integral, also known as the Ito, formally Itô, integral, has the form $\int_a^b f(t, \omega) dB(t, \omega)$, where $f(t, \omega)$ is a stochastic process adapted to the filtration $\mathcal{F}_t = \sigma\{B(s) : s \leq t\}$ and $\int_a^b |f(t)|^2 dt < \infty$. Ito's lemma is the stochastic counterpart of the chain rule in ordinary calculus. Let $f(t, x)$ be a continuous function with continuous partial derivatives $\frac{\partial f}{\partial t}$, $\frac{\partial f}{\partial x}$, and $\frac{\partial^2 f}{\partial x^2}$. Then:

$$f(t, B_t) = f(a, B_a) + \int_a^t \frac{\partial f}{\partial x}(s, B_s) dB_s + \int_a^t \left(\frac{\partial f}{\partial t}(s, B_s) + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(s, B_s) \right) ds. \quad (1)$$

This leads nicely into the definition of an Ito process, where an Ito process is a stochastic process of the form:

$$X_t = X_a + \int_a^t \sigma_s dB_s + \int_a^t \mu_s ds, \quad (2)$$

for $a \leq t \leq b$, where X_a is $\mathcal{F}_a$ -measurable, $\sigma_t \in \mathcal{L}_{ad}(\Omega, L^2([a, b]))$, and $\mu_t \in \mathcal{L}_{ad}(\Omega, L^1([a, b]))$, where $\mathcal{L}_{ad}$ denotes the set of admissible control processes (the subscript, *ad*, stands for admissible, while the script letter, $\mathcal{L}$, represents the collection or space of available control strategies or actions over time). We can rewrite (2) as the following stochastic differential $dX_t = \sigma_t dB_t + \mu_t dt$. Do note that the stochastic differential has no meaning by itself as Brownian paths are nowhere differentiable, it is a symbolic expression to mean the equation (2).

If we let X_t be an Ito process and f a continuous function with continuous partial derivatives, then $f(t, X_t)$ is also an Ito process, where:

$$f(t, X_t) = f(a, X_a) + \int_a^t \frac{\partial f}{\partial x}(s, X_s) dB_s + \int_a^t \left(\frac{\partial f}{\partial t}(s, X_s) + \mu_s \frac{\partial f}{\partial x}(s, X_s) + \frac{1}{2} \sigma_s^2 \frac{\partial^2 f}{\partial x^2}(s, X_s) \right) ds. \quad (3)$$

We can show equation (3) in terms of a stochastic differential. First note that $(dX_t)^2 = \sigma_t^2 (dB_t)^2 + 2\sigma_t \mu_t dB_t dt + \mu_t^2 (dt^2) = \sigma_t^2 dt$. Thus, leading to:

$$\begin{aligned} df(t, X_t) &= \frac{\partial f}{\partial t}(t, X_t) dt + \frac{\partial f}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) (dX_t)^2 \\ &= \sigma_t \frac{\partial f}{\partial x} dB_t + \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt, \end{aligned} \quad (4)$$

where the drift rate is $\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2}$.

STOPPING TIME AND THE REFLECTION PRINCIPLE

We previously introduced stopping time in Article No.29, but intuitively, a stopping time is a moment when a random event related to the process happens. We can now introduce the strong Markov property, stating that for every almost surely finite stopping time, T , the process $\{B(T+t) - B(T) : t \geq 0\}$ is a standard Brownian motion independent of $\mathcal{F}^+(T) = \{A \in \mathcal{F} : A \cap \{T \leq t\} \in \mathcal{F}^+(t) \forall t \geq 0\}$, this is the σ -algebra for every stopping time T .

One interesting application of the strong Markov property is the reflection principle. Let $\{B(t) : t \geq 0\}$ be a standard Brownian motion and T be a stopping time. Then the process, $B(t)\mathbf{1}_{\{t \leq T\}} + (2B(T) - B(t))\mathbf{1}_{\{t \geq T\}}$, which is called the Brownian motion reflected at T , is also a standard Brownian motion.

Thus, as a consequence, if we let $M(t) := \max_{0 \leq s \leq t} B(s)$ and $a > 0$, then $\mathbb{P}(M(t) \geq a) = 2\mathbb{P}(B(t) \geq a) = \mathbb{P}(|B(t)| \geq a)$. There are some incredibly interesting problems which can be solved via the reflection principle.

EXAMPLES USING ITO'S LEMMA

We can use Ito's lemma to determine whether a function is a martingale process or not. Let B_t be a Brownian motion. Is B_t^4 a martingale process?

Solution: We can apply Ito's lemma to $f(t, B_t) = B_t^4$. Here, we have $\frac{\partial f}{\partial t} = 0$, $\frac{\partial f}{\partial B_t} = 4B_t^3$, $\frac{\partial^2 f}{\partial B_t^2} = 12B_t^2$. Thus, from (4), we have $df(t, B_t) = 4\sigma_t B_t^3 dB_t + (4\mu_t B_t^3 + 6\sigma_t^2 B_t^2) dt$. If we take $X_t = B_t$ and look at the standard stochastic differential, then $\mu_t = 0$ and $\sigma_t = 1$. Thus, $df(t, B_t) = 4B_t^3 dB_t + 6B_t^2 dt$. For the cases $B_t \neq 0$, which has probability 1, the drift term, $6B_t^2 dt$, is not zero. Hence B_t^4 is not a martingale process.

Let $B(t)$ be a standard Brownian motion and τ_x , $x > 0$, be the first passage time (the specific moment when a stochastic process hits a boundary/target state for the first time) to level x , $\tau_x = \min\{t, B(t) = x\}$. What is the PDF and the expected value of τ_x ?

Solution: First, let us determine the cumulative distribution function of τ_x . Observe that

$\{\tau_x \leq t\} = \left\{ \sup_{0 \leq s \leq t} B(s) \geq x \right\}$. Thus, $F_{\tau_x}(t) = \mathbb{P}(\tau_x \leq t) = \mathbb{P}\left(\sup_{0 \leq s \leq t} B(s) \geq x\right)$. By the reflection principle, $\mathbb{P}\left(\sup_{0 \leq s \leq t} B(s) \geq x\right) = 2\mathbb{P}(B(t) \geq x)$. Since $B(t) \sim N(0, t)$, we have that $\mathbb{P}(B(t) \geq x) = 1 - \Phi\left(\frac{x}{\sqrt{t}}\right)$, where Φ denotes the standard normal distribution function. Therefore, $F_{\tau_x}(t) = 2\left(1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right)$, $t > 0$. To obtain the density, we differentiate the CDF to obtain: $f_{\tau_x}(t) = \frac{d}{dt}\left[2\left(1 - \Phi\left(\frac{x}{\sqrt{t}}\right)\right)\right]$. Using the chain rule and the standard normal density, that is $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}$, we obtain $f_{\tau_x}(t) = 2\phi\left(\frac{x}{\sqrt{t}}\right) \frac{x}{2t^{\frac{3}{2}}}$. Hence, the PDF of τ_x is $f_{\tau_x}(t) = \frac{x}{\sqrt{2\pi}t^{\frac{3}{2}}} \exp\left(-\frac{x^2}{2t}\right)$, $t > 0$. Do note that this is a common example and is known as the first passage time density of Brownian motion.

Now consider $\mathbb{E}[\tau_x] = \int_0^\infty t f_{\tau_x}(t) dt$. Substituting in the density yields $\mathbb{E}[\tau_x] = \frac{x}{\sqrt{2\pi}} \int_0^\infty t^{-\frac{1}{2}} \exp\left(-\frac{x^2}{2t}\right) dt$. As $t \rightarrow \infty$, $\exp\left(-\frac{x^2}{2t}\right) \rightarrow 1$, and the integrand behaves like $t^{-\frac{1}{2}}$. Since $\int_1^\infty t^{-\frac{1}{2}} dt = \infty$, the integral diverges, and therefore $\mathbb{E}[\tau_x] = \infty$. Thus, although Brownian motion reaches every positive level almost surely, the expected waiting time until it first reaches that level is infinite.

CONCLUSION

Stochastic integrals and Ito's lemma provide the mathematics required to analyse functions of random processes. Together with stopping times and the reflection principle, they can allow us to study important properties of Brownian motion, such as martingales and first passage times.

ALEXANDER FLANN